

The softened-log (Plummer) overlap: a fully analytic mollification

Abstract

The Gaussian mollification of the polygon–polygon overlap area (`smoothOverlap.tex`) is C^∞ but forces Gauss–Legendre quadrature on *both* boundaries, because its potential has no elementary segment integral. Replacing the Gaussian mollifier by the softened-log (Plummer) kernel preserves the smoothness while making every boundary integral tractable in closed form. This note develops the model in three tiers of increasing analyticity: (i) the exact reduction of the mollified overlap and its shape gradient to double boundary integrals; (ii) the closed-form *inner* panel integrals, which remove one quadrature loop ($\mathcal{O}(G^2) \rightarrow \mathcal{O}(G)$ per edge pair, implemented, $11\times$ faster); and (iii) — new here — the closed-form *outer* integrals, which remove the last quadrature ($\mathcal{O}(G) \rightarrow \mathcal{O}(1)$). The final formulas are elementary functions of the four corner–corner softened distances and angles, plus a single transcendental object $\mathcal{T}$ built from four complex dilogarithms per edge pair and per endpoint; the same $\mathcal{T}$ serves the energy and the vertex forces, and the Plummer self-repulsion energy needs no dilogarithms at all. Companion sections treat the operational envelope: quantitative tail size and smooth truncation, kernel selection (rational vs. compactly supported), the sharp $\sigma \rightarrow 0$ limit and its non-uniformities, and a reusable neighbor-list scheme reducing the global cost from $\Theta(N^2 E^2)$ to $\Theta(Nz E^2)$ gates plus $\Theta(Nz E_{\text{act}})$ closed-form evaluations. Every formula below has been verified against adaptive quadrature at 25-digit working precision (`verify_plummer_analytic.py`); [blue text marks implementation status and asides](#).

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1 Overview: why mollify, and the three-tier cost ladder

The sharp pair energy of the deformable-polygon model is built on the exact overlap area $A_\cap = \int_{\mathbb{R}^2} \mathbf{1}_A \mathbf{1}_B \, d^2x$ of two polygons A, B . Exact overlap is piecewise analytic in the vertex positions: its gradient jumps whenever a vertex crosses an edge of the partner, i.e. exactly at contact formation and breaking. Line-search minimizers tolerate this poorly close to a minimum — [sharp CG launched from a well-relaxed configuration hard-stalls at \$\max |F| \approx 3.6 \times 10^{-3}\$](#) — because the energy landscape is only C^0 across the contact set. Mollifying one indicator,

$$A_\cap^\sigma = \int_{\mathbb{R}^2} \mathbf{1}_A (K_\sigma * \mathbf{1}_B) \, d^2x = \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \mathbf{1}_A(\vec{x}) K_\sigma(\vec{x} - \vec{x}') \mathbf{1}_B(\vec{x}') \, d^2x \, d^2x', \quad (1)$$

with a normalized C^∞ kernel K_σ of width σ produces an energy that is C^∞ in all vertex coordinates and converges to $A_\cap$ as $\sigma \rightarrow 0$ [6, 7]. The second form of (1) shows the construction is symmetric in $A \leftrightarrow B$ despite mollifying only one factor: it is a proximity-weighted count of point pairs, one point in each body.

The computational question is what (1) costs. Two divergence-theorem reductions (§3) convert the double area integral into a double *boundary* integral over edge pairs. For the Gaussian kernel neither boundary integral is elementary, so each of the $E_A E_B$ edge pairs costs a $G \times G$ tensor quadrature. The Plummer kernel of §2 makes the inner integral closed form (§5), leaving a single smooth outer integral — the *semi-analytic* tier, implemented in `plummerOverlap.py`. [On the \$N = 6\$ packing this runs at 93 ms per energy+gradient call at outer \$G = 32\$, versus 1040 ms for the Gaussian version: 11×, with force = \$\nabla \text{energy}\$ to \$\sim 10^{-9}\$.](#) Section 6 completes the program: the outer integral also closes, in elementary functions plus dilogarithms, the standard transcendental level for Galerkin double integrations of logarithmic kernels [12, 11, 13]. The ladder is

$$\underbrace{\mathcal{O}(G^2)}_{\text{Gaussian}} \longrightarrow \underbrace{\mathcal{O}(G)}_{\text{Plummer, semi-analytic}} \longrightarrow \underbrace{\mathcal{O}(1)}_{\text{Plummer, fully analytic}}$$

per edge pair. Beyond the constant-factor speedup, the fully analytic tier decouples accuracy from geometry: the semi-analytic outer rule must resolve the $\mathcal{O}(\sigma)$ -scale variation of the integrand where ∂A crosses ∂B , so shrinking σ forces G up; the closed form is exact at any σ . Sections 8 and 10 treat the two remaining axes of cost and fidelity: the $\sigma \rightarrow 0$ limit, and the global pair count.

Context and roadmap. The setting is deformable-particle jamming [19]: each grain is a polygon whose vertices are the degrees of freedom, and the energy sums a shape-holding term (edge and area springs) with the pairwise overlap penalty. Relaxing to mechanical equilibrium and then probing stress and stiffness demands an energy that is both smooth and minimizable to machine precision — neither of which the sharp overlap offers near the jammed contacts, which sit exactly on its C^0 ridge. The mollification (1) buys smoothness at a fixed, controllable bias $\mathcal{O}(\sigma^2)$ (§7), and the reductions below buy speed. The plan: §2 fixes the kernel and its two potentials; §3 runs the two divergence-theorem reductions to boundary integrals; §§4–5 evaluate the inner integrals in closed form (semi-analytic tier); §6 closes the outer integrals (fully analytic tier, the technical heart); and §§7–10 treat tail, truncation, kernel choice, the sharp limit, and the neighbor list. A reader who only wants to *use* the result can take (36)–(40) for the energy and (42)–(43) for the force, with the primitives of §§6.3–6.4.

2 Kernel, potentials, and the softened step

The mollifier is the two-dimensional Plummer (softened point-source) density

$$K_\sigma(\vec{r}) = \frac{\sigma^2}{\pi(|\vec{r}|^2 + \sigma^2)^2}, \quad \int_{\mathbb{R}^2} K_\sigma \, d^2r = 1, \quad (2)$$

a normalized, radially symmetric, C^∞ bump of width σ that tends to $\delta^2(\vec{r})$ as $\sigma \rightarrow 0$. It is the planar analogue of Plummer’s stellar density profile [1] and of the softened point mass of collisionless N -body dynamics [2]; in vortex methods the associated velocity field is exactly Krasny’s algebraically desingularized Biot–Savart kernel [3, 4], and in the blob-kernel classification of Beale and Majda it is the lowest-order rational blob [5].

The boundary reductions of §3 require a vector field $\vec{F}$ with $\nabla \cdot \vec{F} = K_\sigma$ and a scalar Φ with $\nabla \Phi = \vec{F}$. By radial symmetry $\vec{F} = f(r) \hat{r}$ with $\frac{1}{r}(rf)' = K_\sigma$, so

$$f(r) = \frac{1}{r} \int_0^r r' K_\sigma(r') \, dr' = \frac{\sigma^2}{\pi r} \cdot \frac{1}{2} \left[\frac{1}{\sigma^2} - \frac{1}{r^2 + \sigma^2} \right] = \frac{r}{2\pi(r^2 + \sigma^2)},$$

and integrating once more,

$$\vec{F}(\vec{r}) = \frac{\vec{r}}{2\pi(|\vec{r}|^2 + \sigma^2)}, \quad \Phi(r) = \frac{1}{4\pi} \ln(r^2 + \sigma^2). \quad (3)$$

$\vec{F}$ is rational and Φ a *softened logarithm* — no exp, erf or E_1 anywhere, which is the entire point. Both are the point-source pair $\vec{r}/2\pi r^2$, $\frac{1}{2\pi} \ln r$ with the singularity rounded off, and both are smooth at $r = 0$; in particular self-panels and touching panels are perfectly regular, so no singular-quadrature machinery is ever needed. Geometrically $\vec{F}$ is the softened Coulomb field of a unit 2D source: far out it is the bare $\vec{r}/2\pi r^2$ that a point source produces, while within $\sim \sigma$ the $r^2 + \sigma^2$ denominator turns the $1/r$ blow-up into the linear $\vec{r}/2\pi\sigma^2$ that vanishes at the origin. Every boundary integral below is finite precisely because its integrand never sees the point-source singularity — the price the Gaussian pays with exp and E_1 , the Plummer kernel pays with a single extra power in a denominator.

A closed form worth recording is the response to a half-plane. The one-dimensional marginal of (2) is $k_1(x) = \int K_\sigma(x, y) \, dy = \sigma^2 / (2(x^2 + \sigma^2)^{3/2})$, so the mollified indicator of a half-plane at signed distance d from its boundary is

$$s(d) = \int_{-d}^{\infty} k_1 = \frac{1}{2} \left(1 + \frac{d}{\sqrt{d^2 + \sigma^2}} \right) : \quad (4)$$

an *algebraic sigmoid*. The marginal itself is elementary: $k_1(x) = \frac{\sigma^2}{\pi} \int_{-\infty}^{\infty} dy / (x^2 + y^2 + \sigma^2)^2 = \frac{\sigma^2}{\pi} \cdot \frac{\pi}{2(x^2 + \sigma^2)^{3/2}}$ using the standard $\int dy / (A^2 + y^2)^2 = \pi / 2A^3$; integrating from the boundary, $\int_{-d}^{\infty} (x^2 + \sigma^2)^{-3/2} dx = [x / (\sigma^2 \sqrt{x^2 + \sigma^2})]_{-d}^{\infty}$, gives (4). Equation (4) is what the whole construction looks like from one meter away — every sharp indicator in the theory is replaced by this soft step evaluated on a suitable signed distance — and it is odd about its midpoint, $s(d) + s(-d) = 1$, which controls the mollification bias (§7).

The price of an algebraic kernel is the tail. $K_\sigma \sim \sigma^2 / \pi r^4$, so two far, non-overlapping bodies carry a spurious overlap $A_\cap^\sigma \approx \sigma^2 A_A A_B / (\pi r^4)$ that vanishes as $\sigma \rightarrow 0$ but forbids a hard distance cutoff (a cutoff would reintroduce a discontinuity, defeating the construction). §7 quantifies this and sketches faster-decaying rational kernels for which the same analytic program goes through.

3 Exact boundary reductions

Two applications of the divergence theorem, one per potential in (3), peel the two area integrals of (1) down to the boundaries. First, the mollified indicator of B ,

$$\Psi_B(\vec{y}) \equiv (K_\sigma * \mathbf{1}_B)(\vec{y}) = \int_B K_\sigma(\vec{z} - \vec{y}) d^2 z = \oint_{\partial B} \vec{F}(\vec{z} - \vec{y}) \cdot \hat{n}_B(\vec{z}) d\ell_z, \quad (5)$$

using $\nabla_z \cdot \vec{F}(\vec{z} - \vec{y}) = K_\sigma(\vec{z} - \vec{y})$. Then, with $\vec{F}(\vec{z} - \vec{y}) = \nabla_z \Phi = -\nabla_y \Phi$ and the gradient theorem $\int_A \nabla_y \Phi d^2 y = \oint_{\partial A} \Phi \hat{n}_A d\ell_y$,

$$A_\cap^\sigma = \int_A \Psi_B(\vec{y}) d^2 y = - \oint_{\partial A} \oint_{\partial B} \Phi(|\vec{y} - \vec{z}|) (\hat{n}_A \cdot \hat{n}_B) d\ell_z d\ell_y. \quad (6)$$

Because $\partial A, \partial B$ are unions of straight edges e_α, e_β with constant normals $\hat{n}_\alpha, \hat{n}_\beta$, (6) is an *exact* finite sum over edge pairs of double integrals of the softened log $\Phi = \frac{1}{4\pi} \ln(|\vec{y} - \vec{z}|^2 + \sigma^2)$,

$$A_\cap^\sigma = - \sum_{\alpha \in \partial A} \sum_{\beta \in \partial B} (\hat{n}_\alpha \cdot \hat{n}_\beta) \int_{e_\alpha} \int_{e_\beta} \Phi(|\vec{y} - \vec{z}|) d\ell_z d\ell_y. \quad (7)$$

The splitting is exact, not a quadrature; everything below evaluates this single $E_A E_B$ -fold building block in closed form, ending at (40). Both boundaries are traversed counterclockwise with outward normals; since Φ is even in $\vec{y} - \vec{z}$, (6) is manifestly symmetric in $A \leftrightarrow B$. A useful consistency check: because $\oint \hat{n} d\ell = \vec{0}$ on any closed polygon, any *constant* added to Φ drops out of (6); only the variation of the softened log across the configuration contributes (this also explains why far pairs are cancellation-sensitive, §6.9).

Reading the two reductions. Each step trades an area integral for a boundary integral by undoing one potential. The first uses that $\vec{F}$ is a vector potential for the kernel, $\nabla \cdot \vec{F} = K_\sigma$: by the divergence theorem the amount of B seen (softly) from $\vec{y}$, $\Psi_B(\vec{y}) = \int_B K_\sigma$, equals the outward flux of $\vec{F}(\vec{z} - \vec{y})$ through ∂B — a boundary quantity. The second uses that $\vec{F} = \nabla \Phi$ is *itself* a gradient, so $\int_A \Psi_B$ reduces to a flux through ∂A ; carrying both potentials through leaves the symmetric double boundary integral (6). Intuitively Ψ_B is a C^∞ “how much of B do I see” field — 1 deep inside B , 0 far outside, ramping over $\sim \sigma$ across ∂B — and $A_\cap^\sigma = \int_A \Psi_B$ is its average over A , the softened overlap. The gradient (9) is then transparent: moving a vertex sweeps its two incident edges through the field Ψ_B , and the hat weights $(1 - s), s$ apportion each edge’s swept flux to its two endpoints — the same deposit as the sharp force, now reading a soft field rather than a hard indicator.

The shape gradient follows from the transport (Hadamard) theorem [17]: for a boundary velocity field $\vec{V}$ on ∂A ,

$$\delta A_{\cap}^{\sigma} = \oint_{\partial A} \Psi_B(\vec{y}) (\vec{V} \cdot \hat{n}_A) d\ell_y, \quad (8)$$

and symmetrically with Ψ_A on ∂B . With the polygon edge from $\vec{v}_m$ to $\vec{v}_{m+1}$ parametrized affinely, a perturbation $\delta \vec{v}_m$ moves $\vec{y}(s)$ by $(1-s)\delta \vec{v}_m$, so the vertex force rule is

$$\begin{aligned} \frac{\partial A_{\cap}^{\sigma}}{\partial \vec{v}_m} &= L \hat{n}_A \int_0^1 (1-s) \Psi_B(\vec{y}(s)) ds \Big|_{\text{edge } m \rightarrow m+1} \\ &+ L' \hat{n}'_A \int_0^1 s \Psi_B(\vec{y}(s)) ds \Big|_{\text{edge } m-1 \rightarrow m}, \end{aligned} \quad (9)$$

the same $(1-s)$, s hat-function deposit as the sharp force. Equations (6) and (9) are exact; everything that follows is the evaluation of their edge-pair building blocks.

The reduction (6) silently assumes the loops are *simple*: for a self-intersecting polygon the right-hand side computes a winding-weighted area (folded flaps count with winding 0 or 2), so folding can spuriously lower the energy. §9 adds the barrier that keeps loops simple.

Evaluating the edge-pair integral. The building block $\int_{e_{\alpha}} \int_{e_{\beta}} \Phi$ of (7) closes as an *iterated* integral — the inner edge β first, then the outer edge α — with an elementary antiderivative at each stage; the only object beyond elementary functions, the dilogarithm, enters exactly at the second (outer) integration.

Inner edge. Fix the outer point $\vec{y}$ on edge α and parametrize the inner edge by $\vec{z}(t) = \vec{v}_0 + t\vec{e}$, $t \in [0, 1]$, with $\vec{e} = \vec{v}_1 - \vec{v}_0$ and edge length $L_{\beta} = |\vec{e}|$, so the arc-length element is $d\ell_z = L_{\beta} dt$. Since β is straight, the softened squared distance is the quadratic $at^2 + bt + c$ of (13), so the inner integral is a logarithm of a quadratic,

$$\int_{e_{\beta}} \Phi(|\vec{y} - \vec{z}|) d\ell_z = \frac{L_{\beta}}{4\pi} \int_0^1 \ln(at^2 + bt + c) dt. \quad (10)$$

One integration by parts (the log panel (15), derived in §5) evaluates it in closed form as an endpoint *logarithm* $\ln(a+b+c)$ plus the subtended-angle *arctangent* j of (14). Both stay finite for every $\vec{y}$ because $\sigma > 0$ forces the discriminant $D > 0$, even when $\vec{y}$ lies on the edge's line.

Outer edge. Now let $\vec{y} = \vec{y}(s)$ slide along edge α , $s \in [0, 1]$. Edge α is straight too, so the coefficients a, b, c become low-degree *polynomials in* s and the outer integral of (10) is

$$\int_{e_{\alpha}} \int_{e_{\beta}} \Phi d\ell_z d\ell_y = \frac{L_{\alpha} L_{\beta}}{4\pi} \int_0^1 \left[\ln(a+b+c) - 2 + \frac{b}{2a} \ln \frac{a+b+c}{c} + \frac{D}{2a} j \right] ds, \quad (11)$$

an integral of logarithms and arctangents whose arguments are linear in s . This last one-dimensional integral is where the cost tiers part:

- the *semi-analytic* tier evaluates (11) by an order- G Gauss–Legendre quadrature ((16), §5) — the only genuine sum in the scheme, $\mathcal{O}(G)$ per edge pair;
- the *fully analytic* tier uses that $\int \ln \ell(s) ds$ and $\int \arctan \ell(s) ds$ with a linear $\ell(s)$ have antiderivatives in elementary functions and the dilogarithm Li_2 (§6), giving $\mathcal{O}(1)$ per edge pair with no quadrature.

So the double log integral of (7) closes by two antiderivatives: the inner elementary ($\log + \arctan$), the outer elementary $+ \text{Li}_2$. The dilogarithm is not imposed by hand — it is exactly what integrating a logarithm or an arctangent against the *second* linear shape function produces. Summed over edge pairs this is (40); the vertex force (9) uses the same two panels weighted by the hats $(1 - s)$, s .

4 The softened winding number

Ψ_B deserves its own paragraph because it is the geometric heart of the theory. Fix a field point $\vec{y}$ and an edge of B from $\vec{v}_0$ to $\vec{v}_1$, with $\vec{e} = \vec{v}_1 - \vec{v}_0$, $L = |\vec{e}|$, unit tangent $\hat{t} = \vec{e}/L$, outward normal $\hat{n}$, and $\vec{w} = \vec{v}_0 - \vec{y}$. Because the edge tangent is perpendicular to its own normal, $(\vec{z}(t) - \vec{y}) \cdot \hat{n} = \vec{w} \cdot \hat{n}$ is *constant* along the edge — the one-line miracle that makes the inner integrals easy. Writing $w_{\parallel} = \vec{w} \cdot \hat{t}$ and $h^2 = (\vec{w} \cdot \hat{n})^2 + \sigma^2$, the edge contribution to (5) is

$$\psi_{\text{edge}}(\vec{y}) = \frac{1}{2\pi} \frac{\vec{w} \cdot \hat{n}}{h} \left[\arctan \frac{w_{\parallel} + L}{h} - \arctan \frac{w_{\parallel}}{h} \right]. \quad (12)$$

Derivation. Decompose $\vec{z}(t) - \vec{y}$ into its fixed perpendicular part $\vec{w} \cdot \hat{n}$ (constant, by the miracle) and its sliding parallel part $w_{\parallel} + tL$, so $|\vec{z}(t) - \vec{y}|^2 + \sigma^2 = (w_{\parallel} + tL)^2 + h^2$. With $\vec{F} \cdot \hat{n} = (\vec{w} \cdot \hat{n})/2\pi(|\vec{z} - \vec{y}|^2 + \sigma^2)$ and the substitution $u = w_{\parallel} + tL$, the edge contribution to (5) is

$$\psi_{\text{edge}} = \int_0^1 \vec{F}(\vec{z} - \vec{y}) \cdot \hat{n} L dt = \frac{\vec{w} \cdot \hat{n}}{2\pi} \int_{w_{\parallel}}^{w_{\parallel}+L} \frac{du}{u^2 + h^2} = \frac{\vec{w} \cdot \hat{n}}{2\pi h} \left[\arctan \frac{w_{\parallel}+L}{h} - \arctan \frac{w_{\parallel}}{h} \right],$$

the elementary arctangent giving (12). The bracket is the *softened subtended angle* of the edge at $\vec{y}$. As $\sigma \rightarrow 0$, $h \rightarrow |\vec{w} \cdot \hat{n}|$ and (12) is exactly the signed angle the edge subtends at $\vec{y}$, divided by 2π ; summed over the loop this is the winding number. For $\sigma > 0$, Ψ_B is the winding number computed with every perpendicular distance softened, $(\vec{w} \cdot \hat{n})^2 \mapsto (\vec{w} \cdot \hat{n})^2 + \sigma^2$: a C^∞ function equal to 1 deep inside, 0 far outside, and $\frac{1}{2}$ on the boundary (cf. the soft step (4), which (12) reproduces for a single long edge). In algebraic (implementation) variables, with $\vec{z}(t) = \vec{v}_0 + t\vec{e}$ and

$$\begin{aligned} |\vec{z} - \vec{y}|^2 + \sigma^2 &= at^2 + bt + c, \quad a = L^2, \quad b = 2\vec{w} \cdot \vec{e}, \quad c = |\vec{w}|^2 + \sigma^2, \\ D &\equiv 4ac - b^2 = 4(|\vec{e} \times \vec{w}|^2 + L^2\sigma^2) > 0, \end{aligned} \quad (13)$$

one has $\psi_{\text{edge}} = \frac{1}{2\pi} (\vec{w} \cdot \hat{n}) L j$ with

$$j = \int_0^1 \frac{dt}{at^2 + bt + c} = \frac{2}{\sqrt{D}} \left[\arctan \frac{2a+b}{\sqrt{D}} - \arctan \frac{b}{\sqrt{D}} \right]. \quad (14)$$

This is `plummerMeasure`, verified against fine quadrature to $\sim 10^{-15}$ and, in the $\sigma \rightarrow 0$ limit, against the exact indicator on convex and non-convex loops. Note the arctangent arguments in (14) are $(2a+b)/\sqrt{D}$ and $b/\sqrt{D}$; the earlier note carried a typo $(2\sqrt{a}+b)$ — the code was always correct. The strict positivity $D > 0$ for $\sigma > 0$ (even when $\vec{y}$ lies on the line of the edge) is what guarantees that every formula in this document is singularity-free.

5 Closed-form inner panels

Inserting (3) into (6)–(9) and fixing a field point $\vec{y}$, two panel integrals over each edge of B are required: the *arctan panel* (14) for the gradient, and the *log panel* for the energy,

$$\int_0^1 \ln(at^2 + bt + c) dt = \ln(a+b+c) - 2 + \frac{b}{2a} \ln \frac{a+b+c}{c} + \left(2c - \frac{b^2}{2a}\right) j, \quad (15)$$

Plummer mollification: the sharp indicator is replaced by a C^∞ softened one; the overlap becomes a smooth density (no hard edge)

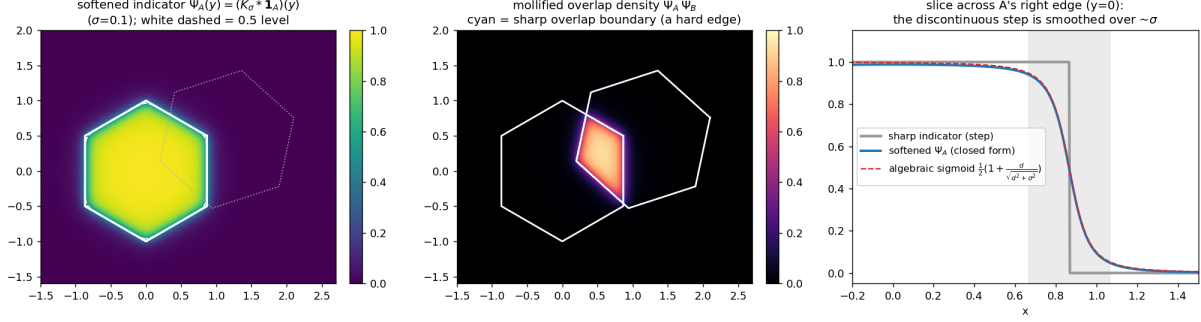


Figure 1: The mollification, computed from the closed forms. *Left:* the softened indicator $\Psi_A = K_\sigma * \mathbf{1}_A$ of one polygon — a C^∞ field, 1 inside, 0 outside, ramping over $\sim \sigma$ (white dashed = the $\frac{1}{2}$ level, which tracks the sharp edge). *Center:* the mollified overlap density $\Psi_A \Psi_B$ of two intersecting polygons is a smooth blob, not a hard-edged lens (cyan = the sharp overlap boundary); its integral is $A_\cap^\sigma$. *Right:* a slice across a flat edge — the sharp indicator step (grey) is replaced by the softened Ψ_A (blue), which lands exactly on the algebraic sigmoid (4) (red dashed), smeared over the $\pm 2\sigma$ band. That replaced step is the entire construction.

with the same j ; note $2c - b^2/2a = D/2a$.

Derivation of the log panel. Integrate by parts with $u = \ln q$ and $dv = dt$ (so $v = t$, $du = q'/q dt$, $q' = 2at + b$):

$$\int_0^1 \ln q dt = [t \ln q]_0^1 - \int_0^1 \frac{t(2at + b)}{q} dt.$$

For the remaining integral, rewrite the numerator against q itself: $2at^2 + bt = 2q - (bt + 2c)$, so $\frac{t(2at+b)}{q} = 2 - \frac{bt+2c}{q}$ and $\int_0^1 t(2at+b)/q dt = 2 - \int_0^1 (bt+2c)/q dt$. Split the last numerator into a piece proportional to q' and a constant, $bt+2c = \frac{b}{2a}(2at+b) + (2c - \frac{b^2}{2a})$; the q' piece integrates to $\frac{b}{2a} \ln q$, and the constant piece, over the completed square $q = a[(t + \frac{b}{2a})^2 + \frac{D}{4a^2}]$, to $(2c - \frac{b^2}{2a}) j$. Assembling with $q(1) = a + b + c$, $q(0) = c$ gives (15). The same two transcendental structures survive all the way to the assembled energy: an endpoint *logarithm* $\ln(a+b+c)$ and a subtended-*angle* arctangent j . These are the classical constant-strength source/doublet panel integrals of two-dimensional potential flow [8, 9]: arctangents are subtended angles, logarithms are endpoint distances, softened here by σ .

Reading the two panels. They are the two things a straight segment does to a field point in planar potential theory. The *arctan panel* j is (a multiple of) the angle the segment subtends — the doublet/vortex influence, and the piece that builds the winding number Ψ_B ; the *log panel* is $\int \ln(\text{distance})$, the source influence that builds the potential Φ and hence the energy. The softening by σ is precisely what a finite core radius does to a point vortex or source. So the gradient reads angles and the energy reads distances — force is a flux of Ψ_B , energy its accumulation. **Closed forms checked against quadrature to 10^{-15} ; the assembled pair energy (`plummerPairEnergy`) matches a fine double-boundary reference to 10^{-16} and the true overlap area as $\sigma \rightarrow 0$.**

With the inner integrals in closed form, the semi-analytic evaluators are

$$\begin{aligned} A_{\cap}^{\sigma} &= -\frac{1}{4\pi} \oint_{\partial A} \left[\sum_{\beta \in \partial B} (\hat{n}_A \cdot \hat{n}_{\beta}) L_{\beta} (\log \text{ panel}) \right] d\ell_y, \\ \frac{\partial A_{\cap}^{\sigma}}{\partial \vec{v}_k} &= \oint_{\partial A} \Psi_B(\vec{y}) (\vec{V} \cdot \hat{n}_A) d\ell_y, \end{aligned} \quad (16)$$

with only the outer ∂A integral quadratured by Gauss–Legendre rules [16] of order G . [Assembly in `plummerOverlap.py`](#): all ordered pairs $A < B$, summed over the 3×3 periodic-image neighborhood — no distance cutoff, the $1/r^4$ tail forbids one. Full-packing force = ∇energy to $\sim 10^{-9}$ at $G = 32$. [Open in `TOD0.md`](#): integer image offsets are square-box only; a faster-decaying rational kernel plus a neighbor list for large N .

6 Fully analytic outer integrals

This section removes the last quadrature. The result: every edge-pair energy and every vertex-force moment is a closed-form expression in the four corner–corner softened distances $r_{ij}^2 = |\vec{v}_i^{\alpha} - \vec{v}_j^{\beta}|^2 + \sigma^2$ and corner angles, built from six elementary primitives and one transcendental object $\mathcal{T}$ (four complex dilogarithms). The structure — and the appearance of Li_2 exactly and only at the double-integration level with linear shape functions — mirrors the analytic-integration literature of the symmetric Galerkin boundary element method [11, 12, 13, 10].

Notation for the edge-pair calculus. These symbols recur throughout §§6.2–6.9:

$\vec{a}, \vec{b}$	edge vectors of $\alpha \subset \partial A$, $\beta \subset \partial B$; L_A, L_B their lengths, $\hat{b} = \vec{b}/L_B$
$\vec{w}_0 = \vec{p}_0 - \vec{q}_0$	offset of the two edge start points
P_0, P_1	along- $\hat{b}$: $p(s) = P_0 + P_1 s$, the foot of $\vec{y}(s)$ on β
X_0, X_1	across- $\hat{b}$: $\chi(s) = X_0 + X_1 s$, the signed height
θ	angle between edges; $X_1 = L_A \sin \theta$, $\alpha = P_1/X_1 = \cot \theta$
$\xi = \chi(s)$	working variable (signed height), $ds = d\xi/X_1$
$U_{0e} = P_0 - eL_B$	start offset to B -endpoint $e \in \{0, 1\}$; $\epsilon_0 = +1$, $\epsilon_1 = -1$
$\beta_e = U_{0e} - \alpha X_0$	endpoint constant in $u_e = \alpha \xi + \beta_e$
$h(s) = \sqrt{\chi^2 + \sigma^2}$	softened perpendicular distance
$Q_2(\xi) = (1 + \alpha^2)\xi^2 + 2\alpha\beta\xi + \beta^2 + \sigma^2$	softened endpoint distance squared
$\Delta = \sqrt{\beta^2 + (1 + \alpha^2)\sigma^2}$	so $4ac - b^2 = 4\Delta^2$ for Q_2
$\Theta = \arctan[(\alpha\xi + \beta)/\sqrt{\xi^2 + \sigma^2}]$	softened endpoint angle
$\zeta_{\pm} = (-\alpha\beta \pm i\Delta)/(1 + \alpha^2)$	complex roots of Q_2
$\eta = \xi + \sqrt{\xi^2 + \sigma^2}$	Euler variable; η_k the four roots of $\mathcal{Q}$

6.1 Warm-up: the sharp kernel by complex corners

For $\sigma = 0$ the double log-panel integral is elementary, by a classical complex trick. Identify $\mathbb{R}^2 \simeq \mathbb{C}$, parametrize the edges as $Y(s) = Y_0 + s\alpha_c$ and $Z(t) = Z_0 + t\beta_c$ with complex slopes α_c, β_c , and note $\ln |\vec{y} - \vec{z}| = \text{Re} \ln(Y - Z)$. Since $H(w) = \frac{1}{4}w^2(2 \ln w - 3)$ satisfies $H'' = \ln w$,

$$\int_0^1 \int_0^1 \ln |\vec{y} - \vec{z}| ds dt = \text{Re} \left[-\frac{1}{\alpha_c \beta_c} \sum_{i,j \in \{0,1\}} (-1)^{i+j} H(Y_i - Z_j) \right], \quad (17)$$

an inclusion–exclusion over the four corner differences (valid with a consistent branch of $\ln$, unproblematic for non-crossing segments).

Why the corners. Check $H'' = \ln w$: $H'(w) = \frac{1}{4}[2w(2\ln w - 3) + 2w] = w \ln w - w$, so $H'' = \ln w$. Now the double integral of any $g(Y - Z)$ affine in (s, t) telescopes. With $g = \ln$ and $G'' = g$ (so $G = H$), integrating t first, $\int_0^1 g(\phi) dt = -\frac{1}{\beta_c}[G'(\phi)]_{t=0}^1$ (since $\partial_t \phi = -\beta_c$), and then s , $\int_0^1 G'(\phi) ds = \frac{1}{\alpha_c}[G(\phi)]_{s=0}^1$ (since $\partial_s \phi = \alpha_c$), gives $-\frac{1}{\alpha_c \beta_c} \sum_{i,j} (-1)^{i+j} G(Y_i - Z_j)$ — a two-fold fundamental theorem of calculus that leaves only the four corner values. Taking the real part is (17). Two lessons carry over to $\sigma > 0$: the answer lives on the corners, and the transcendental level is set by how many antiderivatives of a logarithm are stacked. For $\sigma > 0$ the kernel $\ln(|\vec{y} - \vec{z}|^2 + \sigma^2)$ no longer factors holomorphically — equivalently, it is $2 \ln$ of the *three-dimensional* distance between two skew segments offset by σ — and one extra transcendental level (the dilogarithm) enters, but only in one place.

6.2 Edge-pair coordinates

Fix an ordered edge pair: edge $\alpha \subset \partial A$ from $\vec{p}_0$ with vector $\vec{a}$ ($L_A = |\vec{a}|$), and edge $\beta \subset \partial B$ from $\vec{q}_0$ with vector $\vec{b}$ ($L_B = |\vec{b}|$, $\hat{b} = \vec{b}/L_B$). Decompose everything in the frame of β . With $\vec{w}_0 = \vec{p}_0 - \vec{q}_0$ and the scalar cross product $\vec{u} \times \vec{v} = u_x v_y - u_y v_x$, define the four pair constants

$$P_0 = \vec{w}_0 \cdot \hat{b}, \quad P_1 = \vec{a} \cdot \hat{b}, \quad X_0 = \vec{w}_0 \times \hat{b}, \quad X_1 = \vec{a} \times \hat{b}. \quad (18)$$

Note $P_1^2 + X_1^2 = L_A^2$ (along/perpendicular decomposition of $\vec{a}$). Along $\vec{y}(s) = \vec{p}_0 + s\vec{a}$, the coordinates of the field point relative to β are the along-edge position $p(s) = P_0 + P_1 s$ and the signed perpendicular offset $\chi(s) = X_0 + X_1 s$, both linear in s ; the softened perpendicular distance is $h(s)^2 = \chi^2 + \sigma^2$, and

$$Q(s, t) = |\vec{y}(s) - \vec{z}(t)|^2 + \sigma^2 = (p(s) - L_B t)^2 + \chi(s)^2 + \sigma^2. \quad (19)$$

The two B -endpoints enter through $u_e(s) = p(s) - e L_B$, $e \in \{0, 1\}$, with endpoint signs $\epsilon_0 = +1$, $\epsilon_1 = -1$, and through the softened endpoint distances

$$\begin{aligned} q_e(s) &= u_e^2 + \chi^2 + \sigma^2 = |\vec{y}(s) - \vec{z}_e|^2 + \sigma^2 \\ &= L_A^2 s^2 + 2(U_{0e} P_1 + X_0 X_1)s + (U_{0e}^2 + X_0^2 + \sigma^2), \end{aligned} \quad (20)$$

where $U_{0e} = P_0 - e L_B$. When $X_1 \neq 0$ (non-parallel edges) the substitution $\xi = \chi(s)$, $ds = d\xi/X_1$, turns every s -integral into a ξ -integral in which

$$u_e = \alpha \xi + \beta_e, \quad \alpha = \frac{P_1}{X_1} = \cot \theta, \quad \beta_e = U_{0e} - \alpha X_0, \quad (21)$$

θ being the angle between the edge directions, and the endpoint distance becomes the fixed quadratic

$$Q_2(\xi; \alpha, \beta) = (1 + \alpha^2) \xi^2 + 2\alpha\beta \xi + \beta^2 + \sigma^2 = q_e(s(\xi)) \Big|_{\beta=\beta_e}, \quad Q_2 > 0 \text{ always.} \quad (22)$$

Every non-elementary structure below lives in the pair $(\sqrt{\xi^2 + \sigma^2}, Q_2)$: softened perpendicular distance and softened endpoint distance.

Reading the frame. Everything is written in edge β 's frame: $\hat{b}$ along it, $\hat{n}_\beta$ across. Then P_0, P_1 are the along-edge coordinates (the foot of $\vec{y}(s)$ sits a distance $p(s) = P_0 + P_1 s$ along β), and X_0, X_1 the perpendicular ones ($\chi(s) = X_0 + X_1 s$ is the signed height of $\vec{y}(s)$ above β 's

line). The combinations are geometric: $P_1^2 + X_1^2 = L_A^2$ splits edge A 's length into along/across β ; $X_1 = \vec{a} \times \hat{b} = L_A \sin \theta$ is (up to L_A) the sine of the angle between the edges, so $\alpha = P_1/X_1 = \cot \theta$ diverges exactly as the edges become parallel — which is why $X_1 = 0$ gets its own elementary branch. The substitution $\xi = \chi(s)$ trades the arclength parameter for the signed height above β 's line; it is the natural variable because the softened perpendicular distance is then simply $\sqrt{\xi^2 + \sigma^2}$ and every endpoint distance collapses onto the single quadratic $Q_2(\xi)$.

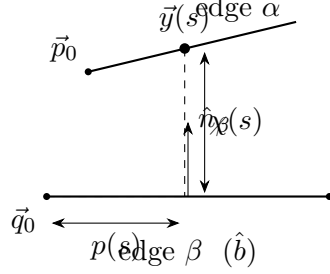


Figure 2: The edge-pair frame (§6.2). In edge β 's frame ($\hat{b}$ along, $\hat{n}_\beta$ across), a field point $\vec{y}(s)$ on edge α has along-edge coordinate $p(s) = P_0 + P_1 s$ and signed height $\chi(s) = X_0 + X_1 s$. The softened perpendicular distance $\sqrt{\chi^2 + \sigma^2}$ and endpoint distance $\sqrt{Q_2}$ carry all the analytic content; $\xi = \chi(s)$ is the working variable.

6.3 Elementary master primitives

Six indefinite integrals, all elementary, carry the load. Throughout, $q(s) = as^2 + bs + c$ with $D = 4ac - b^2 > 0$, $\Theta(\xi) = \arctan[(\alpha\xi + \beta)/\sqrt{\xi^2 + \sigma^2}]$, and $\Delta = \sqrt{\beta^2 + (1+\alpha^2)\sigma^2}$ (so that $4ac - b^2 = 4\Delta^2$ for Q_2).

$$\Lambda_0(s) = \int \ln q \, ds = \left(s + \frac{b}{2a}\right) \ln q - 2s + \frac{\sqrt{D}}{a} \arctan \frac{2as + b}{\sqrt{D}}, \quad (23)$$

$$\Lambda_1(s) = \int s \ln q \, ds = \left(\frac{s^2}{2} - \frac{b^2 - 2ac}{4a^2}\right) \ln q - \frac{s^2}{2} + \frac{bs}{2a} - \frac{b\sqrt{D}}{2a^2} \arctan \frac{2as + b}{\sqrt{D}}, \quad (24)$$

$$\Xi_0(\xi) = \int \frac{d\xi}{Q_2} = \frac{1}{\Delta} \arctan \frac{(1+\alpha^2)\xi + \alpha\beta}{\Delta}, \quad \Xi_1(\xi) = \int \frac{\xi \, d\xi}{Q_2} = \frac{\ln Q_2}{2(1+\alpha^2)} - \frac{\alpha\beta}{1+\alpha^2} \Xi_0, \quad (25)$$

$$R(\xi) = \int \frac{\xi(\alpha\sigma^2 - \beta\xi)}{Q_2} \, d\xi = -\frac{\beta\xi}{1+\alpha^2} + \frac{\alpha[\sigma^2(1+\alpha^2) + 2\beta^2]}{1+\alpha^2} \Xi_1 + \frac{\beta(\beta^2 + \sigma^2)}{1+\alpha^2} \Xi_0, \quad (26)$$

$$M_1(\xi) = \int \frac{\xi}{\sqrt{\xi^2 + \sigma^2}} \Theta \, d\xi = \sqrt{\xi^2 + \sigma^2} \Theta + \frac{\beta}{2(1+\alpha^2)} \ln Q_2 - \frac{\alpha\Delta}{1+\alpha^2} \arctan \frac{(1+\alpha^2)\xi + \alpha\beta}{\Delta}. \quad (27)$$

The derivations are one integration by parts each, using the key derivative

$$\frac{d\Theta}{d\xi} = \frac{\alpha\sigma^2 - \beta\xi}{\sqrt{\xi^2 + \sigma^2} Q_2(\xi)}, \quad (28)$$

whose numerator is linear and whose denominator is exactly the perpendicular–endpoint distance pair: integrating Θ against $d(\sqrt{\xi^2 + \sigma^2}) = \xi \, d\xi / \sqrt{\xi^2 + \sigma^2}$ cancels the square root and leaves a

rational integrand — hence M_1 is elementary. Equation (27) was additionally verified by direct differentiation.

The derivations, worked. Each master is one integration by parts or one algebraic reduction; we record them so every coefficient can be reproduced.

$\Lambda_0 = \int \ln q \, ds$. This is the indefinite log panel: by parts as above, $\int \ln q \, ds = s \ln q - \int s(2as + b)/q \, ds$, and the same numerator substitution $2as^2 + bs = 2q - (bs + 2c)$ with the split $bs + 2c = \frac{b}{2a}(2as + b) + \frac{D}{2a}$ gives $\int s(2as + b)/q \, ds = 2s - \frac{b}{2a} \ln q - \frac{D}{2a} \cdot \frac{2}{\sqrt{D}} \arctan \frac{2as+b}{\sqrt{D}}$. Since $\frac{D}{2a} \cdot \frac{2}{\sqrt{D}} = \frac{\sqrt{D}}{a}$, collecting yields (23). Λ_1 is by parts with $v = s^2/2$, followed by one polynomial division of $s^2(2as+b)$ by q to reduce the integrand to (constant)+(linear)/ q .

$\Xi_0, \Xi_1 = \int \{1, \xi\} \, d\xi/Q_2$. Completing the square, $Q_2 = (1+\alpha^2)[(\xi + \frac{\alpha\beta}{1+\alpha^2})^2 + \frac{\Delta^2}{(1+\alpha^2)^2}]$, exposes the arctangent (25) with scale Δ (this is where $4ac - b^2 = 4\Delta^2$ for Q_2 comes from). For Ξ_1 write $\xi = \frac{1}{2(1+\alpha^2)}[Q_2'(\xi) - 2\alpha\beta]$; the Q_2' part integrates to $\frac{\ln Q_2}{2(1+\alpha^2)}$ and the constant part to $-\frac{\alpha\beta}{1+\alpha^2}\Xi_0$.

$R = \int \xi(\alpha\sigma^2 - \beta\xi)/Q_2 \, d\xi$. Split as $\alpha\sigma^2\Xi_1 - \beta \int \xi^2/Q_2$ and reduce the quadratic numerator with $\xi^2 = \frac{1}{1+\alpha^2}[Q_2 - 2\alpha\beta\xi - (\beta^2 + \sigma^2)]$, so $\int \xi^2/Q_2 = \frac{1}{1+\alpha^2}[\xi - 2\alpha\beta\Xi_1 - (\beta^2 + \sigma^2)\Xi_0]$. Collecting the Ξ_1 coefficient $\alpha\sigma^2 + \frac{2\alpha\beta^2}{1+\alpha^2} = \frac{\alpha[\sigma^2(1+\alpha^2)+2\beta^2]}{1+\alpha^2}$ gives (26).

$M_1 = \int \frac{\xi}{\sqrt{\xi^2 + \sigma^2}} \Theta \, d\xi$ — the one that matters. Integrate by parts against $d\sqrt{\xi^2 + \sigma^2} = \xi \, d\xi/\sqrt{\xi^2 + \sigma^2}$:

$$M_1 = \sqrt{\xi^2 + \sigma^2} \Theta - \int \sqrt{\xi^2 + \sigma^2} \Theta' \, d\xi.$$

The key derivative (28) carries $\sqrt{\xi^2 + \sigma^2}$ in its denominator, so it *cancels* the arc-length factor and the remainder is *rational*, $\int (\alpha\sigma^2 - \beta\xi)/Q_2 \, d\xi = \alpha\sigma^2\Xi_0 - \beta\Xi_1$; substituting Ξ_1 and $\Delta^2 = \beta^2 + (1+\alpha^2)\sigma^2$ yields (27). This single cancellation — the softened perpendicular distance in Θ' matching the boundary arc-length element — is the reason the vertex gradient stays elementary while the energy (which needs one more antiderivative order, §6.4) does not.

6.4 The transcendental core $\mathcal{T}$

One further antiderivative order is needed twice: for the energy ($\int \sqrt{\xi^2 + \sigma^2} \Theta \, d\xi$) and for the linear-weighted gradient ($\int \xi^2(\xi^2 + \sigma^2)^{-1/2} \Theta \, d\xi$). Both integrate by parts through

$$V_{\pm}(\xi) = \frac{1}{2} \left[\xi \sqrt{\xi^2 + \sigma^2} \pm \sigma^2 \operatorname{arcsinh}(\xi/\sigma) \right], \quad V'_+ = \sqrt{\xi^2 + \sigma^2}, \quad V'_- = \frac{\xi^2}{\sqrt{\xi^2 + \sigma^2}}, \quad (29)$$

and the $\xi\sqrt{\cdot}$ halves of $V_{\pm}$ produce the elementary R of (26). The $\operatorname{arcsinh}$ halves produce the single transcendental object of the whole theory,

$$\mathcal{T}(\xi) = \int \operatorname{arcsinh}\left(\frac{\xi}{\sigma}\right) \frac{\alpha\sigma^2 - \beta\xi}{\sqrt{\xi^2 + \sigma^2} Q_2(\xi)} \, d\xi, \quad (30)$$

in terms of which

$$M_2 = \int \sqrt{\xi^2 + \sigma^2} \Theta \, d\xi = V_+ \Theta - \frac{1}{2} R - \frac{\sigma^2}{2} \mathcal{T}, \quad M'_1 = \int \frac{\xi^2 \Theta}{\sqrt{\xi^2 + \sigma^2}} \, d\xi = V_- \Theta - \frac{1}{2} R + \frac{\sigma^2}{2} \mathcal{T}. \quad (31)$$

$\mathcal{T}$ closes in dilogarithms under the Euler substitution

$$\eta = \xi + \sqrt{\xi^2 + \sigma^2} \in (0, \infty), \quad \xi = \frac{\eta^2 - \sigma^2}{2\eta}, \quad \frac{d\xi}{\sqrt{\xi^2 + \sigma^2}} = \frac{d\eta}{\eta}, \quad \operatorname{arcsinh}(\xi/\sigma) = \ln(\eta/\sigma), \quad (32)$$

under which $Q_2 = \mathcal{Q}(\eta)/(4\eta^2)$ with the quartic

$$\mathcal{Q}(\eta) = (1+\alpha^2)(\eta^2-\sigma^2)^2 + 4\alpha\beta\eta(\eta^2-\sigma^2) + 4(\beta^2+\sigma^2)\eta^2, \quad (33)$$

and $\alpha\sigma^2 - \beta\xi = N(\eta)/2\eta$ with $N(\eta) = -\beta\eta^2 + 2\alpha\sigma^2\eta + \beta\sigma^2$. Hence $\mathcal{T} = 2\int \ln(\eta/\sigma) N(\eta)/\mathcal{Q}(\eta) d\eta$; partial fractions over the four simple roots η_k of $\mathcal{Q}$ and the identity $\int \ln \eta (\eta - a)^{-1} d\eta = \ln \eta \ln(1 - \eta/a) + \text{Li}_2(\eta/a)$ give

$$\mathcal{T}(\xi) = 2 \operatorname{Re} \sum_{k=1}^4 \frac{N(\eta_k)}{\mathcal{Q}'(\eta_k)} \left[\ln \frac{\eta}{\sigma} \ln \left(1 - \frac{\eta}{\eta_k} \right) + \text{Li}_2 \left(\frac{\eta}{\eta_k} \right) \right] \quad (34)$$

with $\mathcal{Q}'(\eta) = 4(1+\alpha^2)\eta(\eta^2-\sigma^2) + 4\alpha\beta(3\eta^2-\sigma^2) + 8(\beta^2+\sigma^2)\eta$ and the roots obtained stably from the two conjugate roots of Q_2 ,

$$\zeta_{\pm} = \frac{-\alpha\beta \pm i\Delta}{1+\alpha^2}, \quad \eta \in \{\zeta_{\pm} \pm \sqrt{\zeta_{\pm}^2 + \sigma^2}\} \quad (\text{four values}). \quad (35)$$

Each ζ 's two η -roots multiply to $-\sigma^2$, and the roots come in conjugate pairs, which is why the real part in (34) suffices.

Working the substitution. The Euler map straightens both hard factors at once. From $\eta = \xi + \sqrt{\xi^2 + \sigma^2}$ one has $\eta - \xi = \sqrt{\xi^2 + \sigma^2}$, hence $\eta^2 - 2\eta\xi = \sigma^2$, giving $\xi = (\eta^2 - \sigma^2)/2\eta$ and $\sqrt{\xi^2 + \sigma^2} = (\eta^2 + \sigma^2)/2\eta$; differentiating the first, $d\xi/\sqrt{\xi^2 + \sigma^2} = d\eta/\eta$, and $\operatorname{arcsinh}(\xi/\sigma) = \ln(\eta/\sigma)$. Substituting ξ into Q_2 and clearing the 2η denominators yields $Q_2 = \mathcal{Q}(\eta)/4\eta^2$ with the quartic $\mathcal{Q}$, while the linear numerator becomes $\alpha\sigma^2 - \beta\xi = N(\eta)/2\eta$. The whole integrand then collapses to a logarithm over a quartic:

$$\operatorname{arcsinh} \frac{\xi}{\sigma} \frac{\alpha\sigma^2 - \beta\xi}{\sqrt{\xi^2 + \sigma^2} Q_2} d\xi = \ln \frac{\eta}{\sigma} \cdot \frac{N/2\eta}{\mathcal{Q}/4\eta^2} \cdot \frac{d\eta}{\eta} = 2 \ln \frac{\eta}{\sigma} \frac{N(\eta)}{\mathcal{Q}(\eta)} d\eta.$$

Partial fractions over the four simple roots, $N/\mathcal{Q} = \sum_k \frac{N(\eta_k)}{\mathcal{Q}'(\eta_k)} (\eta - \eta_k)^{-1}$, reduce $\mathcal{T}$ to a sum of $\int \ln \eta (\eta - \eta_k)^{-1} d\eta$ (the $\ln \sigma$ in $\ln(\eta/\sigma)$ only adds an integration constant, cancelled in the definite differences $[\cdot]_{X_0}^{X_0+X_1}$). The remaining antiderivative is the dilogarithm identity

$$\int \frac{\ln \eta}{\eta - a} d\eta = \ln \eta \ln \left(1 - \frac{\eta}{a} \right) + \text{Li}_2 \left(\frac{\eta}{a} \right),$$

checked by differentiating with $\text{Li}_2'(x) = -\ln(1-x)/x$: the two $\ln(1-\eta/a)/\eta$ terms cancel, leaving $-\ln \eta/[a(1-\eta/a)] = \ln \eta/(\eta-a)$. Assembling over the four roots gives (34). *Why one dilogarithm and no more:* the energy stacks two antiderivatives of a logarithm (once in t , once in s), and each $\operatorname{arcsinh}$ half of $V_{\pm}$ is the second stack — so Li_2 enters exactly at the double-integration level, never higher.

Branch bookkeeping. Formula (34) with *principal* branches is globally valid for endpoint differences, with no branch tracking: (i) $\eta > 0$ is real and monotonic in ξ ; (ii) $Q_2 > 0$ forces every η_k off the real axis, so as η sweeps a real interval, η/η_k moves along a straight ray through the origin with non-real direction — it can never meet the Li_2 cut $[1, \infty)$; (iii) $\operatorname{Im}(1 - \eta/\eta_k) = -\eta \operatorname{Im}(1/\eta_k)$ has fixed sign, so the principal $\ln(1 - \eta/\eta_k)$ is continuous along the path; and (iv) integration constants (including the $\ln \sigma$ cross terms) cancel in primitive differences. **Verified: $\mathcal{T}$, M_2 , M_1' against adaptive quadrature over random (α, β, σ) and sign-changing ξ -ranges, agreement at the 10^{-24} level at 25-digit precision.** On dilogarithm identities (inversion, Landen) useful for range reduction and fast evaluation see [14, 15].

Evaluating the dilogarithm. In practice $\text{Li}_2(z)$ on complex z is computed by range reduction to $|z| \leq \frac{1}{2}$, where the series $\text{Li}_2(z) = \sum_{k \geq 1} z^k/k^2$ converges quickly, using the inversion $\text{Li}_2(z) + \text{Li}_2(1/z) = -\frac{\pi^2}{6} - \frac{1}{2} \ln^2(-z)$ for $|z| > 1$ and the reflection $\text{Li}_2(z) + \text{Li}_2(1-z) = \frac{\pi^2}{6} - \ln z \ln(1-z)$ (or the Landen map $\text{Li}_2(z) = -\text{Li}_2(\frac{z}{z-1}) - \frac{1}{2} \ln^2(1-z)$) to pull the argument into that disk [14, 15]. The oracle uses `mpmath.polylog`; the numpy port uses `scipy.special.spence`, related by $\text{Li}_2(z) = \text{spence}(1-z)$ and matched to $\sim 10^{-16}$ against `mpmath` on real and complex arguments; a GPU kernel would inline the reduced series. Because the four roots per pair share $\zeta_{\pm}$, an implementation computes them once and calls Li_2 four times (energy) or eight (energy + gradient) per non-parallel edge pair.

6.5 Assembled energy

The pair integral $I_{\alpha\beta} = \int_0^1 \int_0^1 \ln Q \, ds \, dt$ evaluates by doing t first with the log panel (in the u -variable, $\int \ln(u^2 + h^2) \, du = u \ln(u^2 + h^2) - 2u + 2h \arctan(u/h)$) and then s with the masters:

$$I_{\alpha\beta} = \frac{1}{L_B} \sum_{e \in \{0,1\}} \epsilon_e \left[\mathcal{A}_e - 2\mathcal{B}_e + 2\mathcal{C}_e \right] \quad (36)$$

with, per endpoint e (all bracketed primitives evaluated between the stated limits),

$$\mathcal{A}_e = \int_0^1 u_e \ln q_e \, ds = U_{0e} [\Lambda_0]_0^1 + P_1 [\Lambda_1]_0^1 \quad \text{on } q_e \text{ of (20),} \quad (37)$$

$$\mathcal{B}_e = \int_0^1 u_e \, ds = U_{0e} + \frac{1}{2} P_1, \quad (38)$$

$$\mathcal{C}_e = \int_0^1 h \arctan \frac{u_e}{h} \, ds = \begin{cases} \frac{1}{X_1} \left[M_2(\xi; \alpha, \beta_e) \right]_{X_0}^{X_0+X_1} & X_1 \neq 0, \\ \frac{h}{P_1} \left[u \arctan \frac{u}{h} - \frac{h}{2} \ln(u^2 + h^2) \right]_{U_{0e}}^{U_{0e}+P_1}, & h = \sqrt{X_0^2 + \sigma^2} \quad X_1 = 0, \end{cases} \quad (39)$$

How the pieces arise. Do the inner (t) integral first. In the variable $u = p(s) - L_B t$ (so $Q = u^2 + h^2$ and $dt = -du/L_B$), the log panel is $\int \ln(u^2 + h^2) \, du = u \ln(u^2 + h^2) - 2u + 2h \arctan(u/h)$, evaluated between the two B -endpoints $u_0(s) = p(s)$ and $u_1(s) = p(s) - L_B$. Written as a signed sum over $e \in \{0,1\}$ ($\epsilon_0=+1$, $\epsilon_1=-1$) and divided by the Jacobian L_B , it leaves three outer s -integrals per endpoint. The $u \ln(u^2 + h^2)$ term is $\mathcal{A}_e = \int_0^1 u_e \ln q_e \, ds$ — done by Λ_0, Λ_1 , since $u_e = U_{0e} + P_1 s$ is linear and q_e is the quadratic (20); the $-2u$ term is $-2\mathcal{B}_e$ with $\mathcal{B}_e = \int_0^1 u_e \, ds$, trivial; and the $2h \arctan(u/h)$ term is $2\mathcal{C}_e$ — the *only* place Θ (hence M_2 , hence $\mathcal{T}$) enters. So the endpoint sum (36) is just the two evaluation limits of the inner log panel, propagated through the outer integral.

and the total mollified overlap

$$A_{\cap}^{\sigma} = -\frac{1}{4\pi} \sum_{\alpha \in \partial A} \sum_{\beta \in \partial B} (\hat{n}_{\alpha} \cdot \hat{n}_{\beta}) L_A L_B I_{\alpha\beta}. \quad (40)$$

Parallel edges ($X_1 = 0$, in which case $P_1 = \pm L_A \neq 0$) are *fully elementary*; the non-parallel case costs, per endpoint, one evaluation of (34), i.e. four complex dilogarithms and logarithms at each of the two ξ -limits. Verified: (36) against tensor quadrature of $\ln Q$ over random edge pairs (crossing, disjoint, exactly parallel) to $\sim 10^{-24}$; the full double sum (40) against area quadrature of $\int_A \Psi_B$ for two overlapping quadrilaterals to 4×10^{-25} .

6.6 Assembled gradient

The vertex forces (9) need the hat-weighted moments of Ψ_B along each A -edge. Per B -edge β , writing (12) in pair coordinates ($\vec{w} \cdot \hat{n}_\beta = -\chi$, $w_\parallel = -p$),

$$\psi_\beta(s) = -\frac{\chi(s)}{2\pi h(s)} \sum_e \epsilon_e \arctan \frac{u_e(s)}{h(s)}, \quad (41)$$

so the required moments $W_m = \int_0^1 s^m \psi_\beta ds$ ($m = 0, 1$) are, for $X_1 \neq 0$,

$$\boxed{\begin{aligned} W_0 &= -\frac{1}{2\pi X_1} \sum_e \epsilon_e \left[M_1(\xi; \alpha, \beta_e) \right]_{X_0}^{X_0+X_1}, \\ W_1 &= -\frac{1}{2\pi X_1^2} \sum_e \epsilon_e \left[M'_1(\xi; \alpha, \beta_e) - X_0 M_1(\xi; \alpha, \beta_e) \right]_{X_0}^{X_0+X_1} \end{aligned}} \quad (42)$$

Deriving the moments. Substitute $\xi = \chi(s) = X_0 + X_1 s$ in (41): $s = (\xi - X_0)/X_1$, $ds = d\xi/X_1$, and with $\chi = \xi$, $h = \sqrt{\xi^2 + \sigma^2}$ the integrand becomes (up to $-1/2\pi$) $s^m \frac{\xi}{\sqrt{\xi^2 + \sigma^2}} \sum_e \epsilon_e \Theta(\xi; \alpha, \beta_e)$.

For $m = 0$ this is $\frac{\xi}{\sqrt{\cdot}} \Theta$, whose antiderivative is M_1 — giving W_0 . For $m = 1$, splitting $s = (\xi - X_0)/X_1$ produces $\frac{\xi^2}{\sqrt{\cdot}} \Theta$ and $-X_0 \frac{\xi}{\sqrt{\cdot}} \Theta$, with antiderivatives M'_1 and $-X_0 M_1$ — giving W_1 ; the extra powers of $1/X_1$ are the substitution Jacobians. Since M_1 is elementary and only M'_1 carries $\mathcal{T}$, the net edge force W_0 is elementary and the dilogarithm sits solely in the linear apportionment W_1 , as noted next.

The parallel case is again elementary, using the arctangent primitive of (39) together with

$$\int u \arctan \frac{u}{h} du = \frac{u^2 + h^2}{2} \arctan \frac{u}{h} - \frac{h u}{2}.$$

The force deposit on the edge from $\vec{v}_m$ to $\vec{v}_{m+1}$ is

$$\frac{\partial A_\cap^\sigma}{\partial \vec{v}_m} += L_A \hat{n}_A \sum_{\beta \in \partial B} (W_0 - W_1), \quad \frac{\partial A_\cap^\sigma}{\partial \vec{v}_{m+1}} += L_A \hat{n}_A \sum_{\beta \in \partial B} W_1, \quad (43)$$

plus the mirror-image contributions on ∂B with Ψ_A . Two structural remarks. First, W_0 — the *net* force integral of an edge — is purely elementary: dilogarithms enter only through the linear apportionment W_1 , i.e. exactly at the piecewise-linear (Galerkin) level, in agreement with the symmetric Galerkin analytic-integration literature [11, 12]. Second, W_0, W_1 reuse the same (α, β_e) and root data as the energy, so an implementation computes $\zeta_\pm, \eta_k$ once per edge pair. **Verified:** (42) against quadrature of $s^m \psi_\beta$ to $\sim 10^{-23}$, and the fully assembled $\partial A_\cap^\sigma / \partial \vec{v}$ (all vertices, both polygons) against central finite differences of the closed-form energy to 1×10^{-13} — FD-limited.

6.7 Worked case: parallel edges

The parallel branch ($X_1 = 0$) is worth carrying through in full: it is the elementary skeleton, and near-aligned edges are generic at jamming, so a production code runs it often. Here $\chi(s) \equiv X_0$ is a fixed signed gap, $h = \sqrt{X_0^2 + \sigma^2}$ is constant, and $P_1 = \pm L_A$; take $P_1 = L_A$ (co-oriented edges). The endpoint abscissa is $u_e(s) = U_{0e} + L_A s$ with $U_{0e} = P_0 - e L_B$, and $q_e = u_e^2 + h^2$. Changing to $u = u_e$ ($ds = du/L_A$), each master collapses to a standard endpoint evaluation:

$$\mathcal{A}_e = \frac{1}{L_A} \left[\frac{1}{2} (u^2 + h^2) \ln(u^2 + h^2) - \frac{1}{2} u^2 \right]_{U_{0e}}^{U_{0e} + L_A}, \quad \mathcal{B}_e = U_{0e} + \frac{1}{2} L_A, \quad \mathcal{C}_e = \frac{h}{L_A} \left[u \arctan \frac{u}{h} - \frac{h}{2} \ln(u^2 + h^2) \right]_{U_{0e}}^{U_{0e} + L_A},$$

using $\int u \ln(u^2+h^2) du = \frac{1}{2}(u^2+h^2) \ln(u^2+h^2) - \frac{1}{2}u^2$ (mod a constant that cancels in the difference) and the arctangent primitive of (39). There is no $\mathcal{T}$, no dilogarithm, no complex arithmetic: the whole edge-pair energy $I_{\alpha\beta} = L_B^{-1} \sum_e \epsilon_e [\mathcal{A}_e - 2\mathcal{B}_e + 2\mathcal{C}_e]$ is four logarithms and four arctangents of the corner distances and angles, exactly the σ -softened version of the sharp parallel-strip overlap. Geometrically it depends only on the perpendicular gap (through h) and on how far the two edges overlap when projected onto their common direction (through the U_{0e}); as $\sigma \rightarrow 0$, $h \rightarrow |X_0|$ and it reduces to the elementary area of the overlapping parallel strip. This is the branch the near-parallel limit (§6.9) is matched to.

6.8 Consistency checks the closed forms satisfy

Four invariances double as correctness checks on an implementation. *Translation invariance*: rigidly translating a polygon leaves every pairwise overlap unchanged, so the net overlap force on it vanishes, $\sum_m \partial A_\cap^\sigma / \partial \vec{v}_m = \vec{0}$ — the two hat deposits (43) must cancel across a polygon; verified to $\sim 10^{-15}$. *Newton’s third law*: since (6) is symmetric in $A \leftrightarrow B$, the force a translation of A feels from B is minus that B feels from A , so momentum is conserved pairwise. *Gauge*: adding a constant to Φ changes nothing, by $\oint \hat{n} d\ell = \vec{0}$, so only *differences* of the softened log across a pair contribute — the origin of the far-pair cancellation (§6.9). *Sharp limit*: $\sigma \rightarrow 0$ collapses every formula onto the elementary complex-corner skeleton §6.1, the explicit σ^2 prefactors on the $\mathcal{T}$ sector switching the dilogarithms off (§8). The first two are the cheapest live guards against a sign or index slip, and a production code should assert them at every step.

6.9 Degenerate limits, conditioning, and cost

Near-parallel edges. As $X_1 \rightarrow 0$, $\alpha = \cot \theta \rightarrow \infty$ and (42)–(39) become a difference of large quantities. The formulas remain exact; a float64 production code should nevertheless switch to the elementary parallel branch below a threshold $|X_1| \lesssim \varepsilon L_A$ and, if needed, bridge with a first-order expansion in X_1 — the standard panel-method treatment [9]. The instability is superficial: each moment is a difference quotient in the limits X_0 and $X_0 + X_1$ (e.g. $\mathcal{C}_e, W_0 \sim [F]_{X_0}^{X_0+X_1}/X_1$), exact for all X_1 but cancellation-prone as $X_1 \rightarrow 0$; its limit is the derivative $F'(X_0)$, which *is* the parallel-branch formula (a segment at fixed height X_0). So the two branches meet smoothly, and a first-order Taylor expansion of the difference quotients removes the cancellation analytically if a single code path is wanted. *Far pairs*. By $\oint \hat{n} d\ell = 0$ the constant part of the log cancels between edges, so distant pairs are computed as small differences of $\mathcal{O}(\ln r)$ terms; evaluating pairs in a local frame centered on the pair and compensated summation keep this benign; beyond the switch radius (§7.2) far pairs are absent altogether. *Root degeneracies*. The η_k are simple and nonzero for all $\sigma > 0$ except at isolated parameter points ($\alpha = \beta = 0$, where $\Theta \equiv 0$ and $\mathcal{T}$ is not needed); a guard suffices.

Operation count. Per non-parallel edge pair, the energy costs $2 \times (34) = 8$ complex dilogarithms + 8 complex logs, plus $\mathcal{O}(10)$ real $\ln/\arctan/\sqrt{\cdot}/\operatorname{arcsinh}$; the gradient adds the same $\mathcal{T}$ data plus a handful of elementary terms. The semi-analytic tier at $G = 32$ spends roughly $32 \times$ a few real transcendentals per pair, so the closed form is an order of magnitude fewer transcendental evaluations — [consistent with the observed 11× from removing the previous quadrature level](#) — with the caveat that complex arithmetic and Li_2 evaluations (series plus inversion/Landen reductions [14]; `mpmath/scipy` host side) are individually more expensive than a real $\arctan$. The qualitative win is accuracy: machine-exact independent of σ and of how the boundaries cross, so σ can be pushed down without raising any quadrature order. [Status: the fully analytic tier is derived and verified end-to-end in `verify-plummer-analytic.py` \(this document’s companion\); the CUDA](#)

port in `pyCudaPolygon` is pending, with the complex- Li_2 kernel the only new device primitive.

7 Tail, truncation, and kernel selection

Three error mechanisms separate $A_\cap^\sigma$ from the true overlap. (i) *Far-pair tail*: for disjoint bodies at separation $r \gg \sigma$, expanding (1) gives $A_\cap^\sigma \approx A_A A_B K_\sigma(r) = \sigma^2 A_A A_B / (\pi r^4)$ — spurious but smooth, vanishing as σ^2 ; it is the price of forbidding cutoffs. (ii) *Straight-boundary crossings*: where ∂B passes through the interior of A , the soft step (4) is odd about $\frac{1}{2}$, so the leading-order error integrates to zero along straight portions; net contributions arise only within $\mathcal{O}(\sigma)$ of $\partial A \cap \partial B$ intersection points, each contributing $\mathcal{O}(\sigma^2)$. (iii) *Corners*: each corner of one polygon inside the other contributes $\mathcal{O}(\sigma^2)$ with an angle-dependent coefficient. A caution on sharpening these statements: the Plummer kernel’s radial second moment $\int r^2 K_\sigma \, d^2r$ diverges logarithmically, so the usual moment-condition bookkeeping of blob kernels [5] does not apply verbatim, and logarithmic corrections to the σ^2 scalings should not surprise. Empirically the pair energy converges to the exact overlap as $\sigma \rightarrow 0$ (checked in `plummerPairEnergy`); a measured convergence-order plot is still to be produced.

7.1 How bad is the tail, quantitatively?

Normalize the rational family as $K_\sigma^{(m)} = (m/\pi) \sigma^{2m} / (r^2 + \sigma^2)^{m+1}$, so $m=1$ is (2). For two disjoint bodies at center separation r large compared with their sizes,

$$E_{\text{tail}}(r) \simeq \frac{m}{\pi} \frac{\sigma^{2m} A_A A_B}{r^{2m+2}}, \quad F_{\text{tail}}(r) \simeq \frac{(2m+2)m}{\pi} \frac{\sigma^{2m} A_A A_B}{r^{2m+3}}, \quad (44)$$

These follow by factorization: at center separation r much larger than either body the kernel is nearly constant across each, so (1) gives $A_\cap^\sigma \approx A_A A_B K_\sigma^{(m)}(r)$, and $K_\sigma^{(m)}(r) \approx (m/\pi) \sigma^{2m} r^{-2m-2}$ for $r \gg \sigma$ is E_{tail} ; the force $-dE_{\text{tail}}/dr$ pulls down the extra $(2m+2)/r$. So matching the *untruncated* model to a force tolerance ε costs a cutoff radius

$$r_c(\varepsilon) = \left[\frac{(2m+2)m \sigma^{2m} A_A A_B}{\pi \varepsilon} \right]^{1/(2m+3)} \sim \varepsilon^{-1/(2m+3)}. \quad (45)$$

For Plummer ($m=1$) at $\sigma = 0.05$, $A_A \sim A_B \sim 1$, $\varepsilon = 10^{-12}$: $r_c \approx 80$ — no useful cutoff exists; the $m=3$ member’s σ^6/r^8 tail brings $r_c \approx 4$. The $\varepsilon^{-1/5}$ scaling is the quantitative content of “the tail is bad”. It is also the wrong requirement, as follows.

7.2 Smooth truncation

“No cutoff” forbids only *discontinuous* truncation. Multiply each edge-pair energy by a smooth switch of the midpoint separation $r_{\alpha\beta} = |\vec{m}_\alpha - \vec{m}_\beta|$,

$$E_{\alpha\beta} \mapsto S(r_{\alpha\beta}) E_{\alpha\beta}, \quad S \equiv 1 \text{ for } r \leq R_{\text{on}}^{\alpha\beta}, \quad S \equiv 0 \text{ for } r \geq R_{\text{off}}^{\alpha\beta}, \quad (46)$$

with $R_{\text{on/off}}^{\alpha\beta} = \frac{1}{2}(L_\alpha + L_\beta) + g_{\text{on/off}}$ and a C^∞ bridge between (a standard bump partition; a C^2 quintic smoothstep suffices for FIRE/CG if a transcendental-free switch is preferred). The switched model is exactly as smooth as the unswitched one, keeps force = ∇energy *exactly* — the product-rule term deposits $\pm \frac{1}{2} S' E_{\alpha\beta} \hat{r}$ on the four endpoints, since $\partial \vec{m} / \partial \vec{v} = \frac{1}{2}$ — and, with g_{on} a few σ beyond genuine contact range, differs from untruncated Plummer only where the interaction is pure spurious tail: a modification of the same $\mathcal{O}(\sigma^{2m})$ size as the mollification bias above, vanishing

in the sharp limit exactly as the bias does. The switch is therefore part of the model definition, not an approximation to be ε -matched; (45) prices the stricter requirement one is declining. Its operational payoff: $S \equiv 0$ beyond R_{off} makes far edge pairs *identically* zero, licensing exact gates and reusable neighbor lists (§10). [Recommended first move: no kernel change, no new analytics — the closed forms of §6 are untouched; the switch multiplies them.](#)

7.3 Kernel selection

Within full-support radial kernels, nothing cleaner than the softened log exists. Writing $\vec{F} = g(u) \vec{r}$ with $u = |\vec{r}|^2$ gives $K = 2 \, d(ug)/du$, so normalization forces $ug \rightarrow 1/2\pi$ at large u and regularity forces $ug \rightarrow 0$ at $u = 0$; the minimal rational interpolant with a single scale is $ug = u/[2\pi(u + \sigma^2)]$ — which *is* Plummer, with $\Phi = \frac{1}{2} \int g \, du = \ln(u + \sigma^2)/4\pi$. The logarithm, and with it the dilogarithm core $\mathcal{T}$, is structural: every normalized member of the family carries the same $\ln$ coefficient $1/4\pi$, plus rational corrections.

The derivation template is not tied to (2). Any radial rational kernel, e.g. the higher-order family $K_\sigma^{(m)} \propto \sigma^{2m}/(r^2 + \sigma^2)^{m+1}$ with $m \geq 2$, has finite second moment and a faster tail $\sim r^{-2(m+1)}$, permitting neighbor lists in practice while retaining C^∞ . The radial ODE gives $\vec{F}$ rational and Φ equal to the same softened log plus a rational correction, so the panel algebra of §§5–6 carries over with additional *elementary* terms only ($\mathcal{T}$ is unchanged in kind). [This is the concrete route for the TODO.md item “faster-decaying rational kernel + neighbor list for large \$N\$ ”.](#)

Two neighbors outside the rational family are worth recording. The Cauchy–Poisson kernel $K = (\sigma/2\pi)(r^2 + \sigma^2)^{-3/2}$ has arguably the prettiest potential and step,

$$\Phi = \frac{1}{2\pi} \ln(\sigma + \sqrt{r^2 + \sigma^2}), \quad s(d) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{d}{\sigma}, \quad (47)$$

but a slower σ/r^3 tail, and because Φ mixes r through $\sqrt{r^2 + \sigma^2}$ its double-panel integrals are no simpler. The genuinely different option is a *compactly supported* polynomial kernel $K_p \propto (1 - r^2/\sigma^2)_+^p$ (Wendland type [25]). By the two-dimensional shell theorem, $\Phi = \frac{1}{2\pi} \ln r + \text{const}$ *exactly* for $r \geq \sigma$ (the constant immaterial by $\oint \hat{n} \, d\ell = 0$) and an even polynomial of degree $2p+2$ inside; Ψ_B is identically 0 or 1 outside a σ -collar of ∂B , and the pair energy *vanishes identically* for gaps $> \sigma$: exact neighbor lists with no switch, no tail, no image subtleties. Edge pairs with clearance $> \sigma$ use the sharp complex-corner formula (17) verbatim; near pairs reduce to polynomial double integrals over the clipped region $Q \leq \sigma^2$ (an ellipse–unit-square intersection) plus the corner formula on the remainder — elementary throughout, no Li_2 , but piecewise, and only finitely smooth: $K_p \in C^{p-1}$ at the support edge gives an energy of class $\approx C^{p+1}$ across branch boundaries, ample for FIRE/CG at $p = 3\text{--}4$. Trade summary: Plummer = uniform branch-free formulas, one transcendental, algebraic tail managed by the switch; compact polynomial = fully elementary with the best possible complexity, at the price of clipping case analysis and finite smoothness. [Default remains Plummer + switch; revisit the compact kernel if complex \$\text{Li}_2\$ proves awkward on the GPU.](#)

8 The sharp limit $\sigma \rightarrow 0$

By construction the model degenerates to the sharp one: $K_\sigma \rightarrow \delta^2$, the soft step (4) $\rightarrow$ the Heaviside step, $\Psi_B \rightarrow$ the winding number (each edge’s arctangent pair in (12) becomes the true subtended angle), and $A_\cap^\sigma \rightarrow A_\cap$ with the $\mathcal{O}(\sigma^2)$ -up-to-logs bias of §7. The limit is also visible *inside* the closed forms, and constitutes a nontrivial consistency check: every transcendental piece carries an explicit σ^2 prefactor — the arcsinh halves of $V_\pm$ and the entire $\mathcal{T}$ sector enter as $\frac{\sigma^2}{2} \mathcal{T}$ in (31) — while $\mathcal{T}$

grows at most like $\ln^2(1/\sigma)$, so the dilogarithm content of the theory switches off as $\sigma^2 \ln^2(1/\sigma)$ and the surviving elementary terms must (and do) reassemble into the sharp complex-corner formula (17). The dilogarithms are genuinely finite- σ objects.

The switch-off, in detail. In (31) the entire dilogarithm sector enters only as $\pm \frac{\sigma^2}{2} \mathcal{T}$. As $\sigma \rightarrow 0$, $\mathcal{T}$ grows at most like $\ln^2(1/\sigma)$ — its dilogarithm arguments stay bounded except near the small roots $\eta \approx -\sigma^2/2\zeta_{\pm}$ of (35), which supply the $\ln^2$ — so $\frac{\sigma^2}{2} \mathcal{T} = \mathcal{O}(\sigma^2 \ln^2(1/\sigma)) \rightarrow 0$. The surviving $V_{\pm}\Theta - \frac{1}{2}R$ are elementary, and Θ becomes the true sharp corner angle $\arctan[(\alpha\xi + \beta)/|\xi|]$; they must and do reassemble into the complex-corner formula (17). This reassembly is a nontrivial check that the integration constants and branch choices of §6.4 are correct. Physically, the softening is a finite- σ deformation of a genuinely elementary $\sigma = 0$ answer: the dilogarithm is the price of a finite contact width, not of the geometry.

The convergence is deliberately not uniform in derivatives. The pair-force crossover at a contact compresses into a window of width σ , so landscape curvature grows as $1/\sigma$ and the stable explicit-integrator step (FIRE is velocity-Verlet based [18]) shrinks as $\sqrt{\sigma}$; at small enough σ one has faithfully recovered the sharp model *including* the hard-stall pathology the mollification exists to escape. Nor should the formulas be evaluated *at* $\sigma = 0$: self- and collinear panels lose $D > 0$ in (13), crossing pairs reacquire the logarithmic singularity, and the small roots $\eta \approx -\sigma^2/2\zeta_{\pm}$ of (35) drive Li_2 arguments to size σ^{-2} , requiring the inversion identity [14] and care in float64 — conditioning degrades before the mathematics does.

Since the analytic tier's cost is σ -independent (§6.9), the natural protocol is σ -continuation: relax at generous σ , anneal downward, and stop when the $\mathcal{O}(\sigma^2)$ bias meets the force or observable tolerance, rather than at the limit itself. *Not yet scripted; the $N = 6$ endpoint of §9 used fixed $\sigma = 0.05$.*

9 Self-repulsion (self-avoidance)

The mollified minimum is meaningful only on *simple* loops: a self-intersecting polygon has winding $\notin \{0, 1\}$, the reduction (6) then computes a winding-weighted area, and folding a flap spuriously lowers the energy — both FIRE [18] and CG dive into it if allowed. `selfRepulsion.py` adds a smooth barrier between every non-adjacent edge pair of a polygon,

$$U_{\text{self}} = \frac{k_{\text{self}}}{2} \sum_{\text{polygons non-adj}} \sum_{(i,j)} \int_0^1 \int_0^1 \exp\left[-\frac{|\vec{y}(s) - \vec{z}(t)|^2}{2\delta^2}\right] ds dt, \quad (48)$$

which is ≈ 0 when non-adjacent edges are $\gtrsim \delta$ apart and rises to $\sim k_{\text{self}}$ as they approach, blocking the crossing. *Gaussian bump, quadratured (few edge pairs per polygon, cheap); FD force = $-\nabla U$ to 6×10^{-11} . First cut — form, k_{self} , δ still to be pinned.* Conceptually this is the polygonal cousin of self-avoidance energies for curves [20, 21].

The Plummer machinery closes this integral too, and — pleasantly — without dilogarithms. Replacing the Gaussian bump by the matched rational bump δ^4/Q^2 with $Q = |\vec{y} - \vec{z}|^2 + \delta^2$ (maximum 1 at contact, tail δ^4/r^4), the required double integral is

$$S = \int_0^1 \int_0^1 \frac{ds dt}{Q(s, t)^2} = \frac{1}{2L_B X_1} \sum_e \epsilon_e \int_{X_0}^{X_0 + X_1} \left[\frac{\alpha\xi + \beta_e}{(\xi^2 + \delta^2) Q_2} + \frac{\Theta_e}{(\xi^2 + \delta^2)^{3/2}} \right] d\xi, \quad (49)$$

(inner t -integral by the standard $\int dt/q^2$ reduction; $\sigma \rightarrow \delta$ throughout in Q_2 , Θ , $\zeta_{\pm}$), where the first term is rational — residues over the four poles $\{\pm i\delta, \zeta_{\pm}\}$, summed as $\text{Re} \sum_k r_k \ln(\xi - \xi_k)$ —

and the second closes by parts,

$$\int \frac{\Theta \, d\xi}{(\xi^2 + \delta^2)^{3/2}} = \frac{\xi \Theta}{\delta^2 \sqrt{\xi^2 + \delta^2}} - \frac{1}{\delta^2} \int \frac{\xi(\alpha\delta^2 - \beta\xi)}{(\xi^2 + \delta^2) Q_2} d\xi, \quad (50)$$

whose remainder is again rational: *fully elementary*, parallel case included; retaining the prefactor convention of (48), each non-adjacent pair contributes $\frac{k_{\text{self}}}{2} \delta^4 S$. The vertex gradient follows by differentiating the closed form with respect to the four endpoints (mechanical; no new transcendentals). Equation (49) verified against tensor quadrature to 3×10^{-22} . Swapping (48)’s Gaussian for the rational bump would make the barrier quadrature-free and C^∞ with an algebraic tail — consistent with the overlap kernel — at the cost of the same no-cutoff constraint; whether the δ^4/r^4 tail between distant edges of a large floppy polygon matters is a question for the k_{self}, δ pinning pass.

The inner reduction, worked. The barrier’s inner integral is the standard $\int_0^1 dt/Q^2$. In $u = p(s) - L_B t$ (so $Q = u^2 + h^2$),

$$\int \frac{du}{(u^2 + h^2)^2} = \frac{u}{2h^2(u^2 + h^2)} + \frac{1}{2h^3} \arctan \frac{u}{h},$$

so evaluating between the two B -endpoints and passing to $\xi = \chi(s)$ ($h^2 = \xi^2 + \delta^2$, $u^2 + h^2 = Q_2$) yields exactly the two s -integrands of (49): a rational one, and $\Theta/(\xi^2 + \delta^2)^{3/2}$. The rational s -integral is a sum of residues over the four poles $\{\pm i\delta, \zeta_\pm\}$ — the softened-perpendicular pair $\pm i\delta$ and the endpoint pair $\zeta_\pm$ — and the second closes by parts (50) into another rational integral. No Θ -primitive of the $M_2/\mathcal{T}$ kind is needed, so the barrier is dilogarithm-free even though it reuses the same $Q_2, \zeta_\pm$ data as the overlap: the extra power in Q^2 (versus $\ln Q$) trades the transcendental for a higher-order rational.

End-to-end status. FIRE on $A_\cap^\sigma + U_{\text{eqSoftBody}} + U_{\text{self}}$ ($N = 6$, $\sigma = \delta = 0.05$, $k = 1$) relaxes to $\max |F| = 9.9 \times 10^{-9}$ in 2159 steps (~ 3 min), simple throughout — the machine-precision, fold-free minimization the sharp model cannot reach (its gradient jumps at contact; sharp CG from this minimum hard-stalls at $\max |F| \approx 3.6 \times 10^{-3}$).

10 Neighbor lists, images, and global complexity

With the switch of §7.2 (or a compact kernel, §7.3) far pairs are identically zero, and the current all-pairs-times-3×3-images sum — $\Theta(N^2 E^2)$ per force call for N polygons of E edges — can be replaced by a reusable Verlet list [22, 23] over polygon bounding disks, built with a cell grid and refined by per-edge-pair gates. What makes reuse clean for *deformable* bodies is a one-line covering lemma: every boundary point is a convex combination of its edge’s endpoints, $\vec{y}(s) = (1-s)\vec{v}_i + s\vec{v}_{i+1}$, so if no vertex has moved more than d since a reference configuration, then every boundary point, every edge midpoint, and the vertex-mean center have moved by at most d . Stretch, rotation, and growth of the bounding radius are covered automatically; a single scalar — the maximum accumulated vertex displacement d_{max} — governs list validity, with no shape-dependent terms.

Polygon Verlet list. At build time compute the vertex mean $\vec{c}_P$ and covering radius $\rho_P = \max_i |\vec{v}_i - \vec{c}_P|$, *measured fresh* at each rebuild (for the record $\rho_P \leq \text{perimeter}/4$, the doubled-segment extremal, so the soft perimeter energy bounds it a priori). List all pairs with minimum-image

$$|\vec{c}_A - \vec{c}_B| \leq \rho_A + \rho_B + R_{\text{int}} + s, \quad R_{\text{int}} = \frac{1}{2}(L_{\text{max}}^A + L_{\text{max}}^B) + g_{\text{off}}, \quad (51)$$

s the skin. Every excluded pair has all edge-pair midpoint separations above their R_{off} and contributes exactly zero. The list is valid while $2d_{\text{max}} < s$: the per-vertex rebuild trigger is a displacement of $s/2$, exactly as for point particles. The σ - and edge-length dependence lives entirely in the static radius (51); measured covering radii remove the explicit L -padding a vertex-ball list needs (two edges can close mid-span while every vertex pair stays $\mathcal{O}(L)$ apart). A sharper trigger: rebuild when the two largest per-polygon displacements sum to s .

Athermal quasistatic shear. Affine strain erodes (51) at a rate proportional to separation: rebuild when $2d_{\text{max}}^{\text{na}} + \Delta\gamma_{\text{acc}} R_{\text{list}} \geq s$, with d^{na} the accumulated non-affine displacement, $\Delta\gamma_{\text{acc}}$ the strain since build, and R_{list} the largest listing radius. Storing positions in lattice (box) coordinates makes Lees–Edwards imaging [24] a remap and confines all strain dependence to the box matrix; the erosion term is metric-intrinsic and remains either way. [This supersedes the square-box-only integer image offsets of TODO.md.](#)

Cells. Bin centers on a grid of cell edge $\geq 2\rho_{\text{max}} + R_{\text{int}} + s$ (in lattice coordinates), 3×3 stencil, counting sort on the GPU. Cells accelerate *builds* only; a pure cell scheme re-traverses every step and re-shifts stencils at each strain increment. The Verlet list is the reuse structure.

Edge gate. Inside each listed pair, test each edge pair by $|\vec{m}_\alpha - \vec{m}_\beta| \leq \frac{1}{2}(L_\alpha + L_\beta) + g_{\text{off}}$, with midpoints computed once per call — exact, because a segment lies inside its midpoint disk of radius $L/2$ and the switch vanishes beyond R_{off} . A gate costs a few flops against $\sim 10^2$ transcendentals per surviving closed-form evaluation, negligible at small E . The identical gate at range δ prunes intra-polygon self-repulsion pairs (§9) if E grows. A flat edge-disk list (midpoints as the binned primitive, identical invalidation rule) wins only for strongly elongated shapes or E of many tens, when circumdisks over-cover badly.

Sizing, cadence, complexity. Be generous with the skin — $s \approx g_{\text{off}}$ — since false candidates cost gates, not dilogarithms. Rebuild cadence is $(s/2)/\langle d_{\text{max}} \text{ per step} \rangle$; FIRE displacements collapse near the endpoint, so the list freezes precisely where long relaxations spend their steps. Per force call the cost is $\Theta(NzE^2)$ gates plus $\Theta(NzE_{\text{act}})$ closed-form evaluations, with $z \approx 6$ –10 listed neighbors for compact shapes near jamming and E_{act} the gate survivors (a few per touching pair); builds are $\Theta(N)$ amortized. The 3×3 image sum collapses to minimum image on listed pairs; explicit self-image pairs are needed only if the box shrinks below $2\rho_{\text{max}} + R_{\text{int}} + s$.

11 Validation summary

Semi-analytic tier (implemented): [inner panels vs. fine quadrature \$\sim 10^{-15}\$](#) ; [pair energy vs. double-boundary reference \$10^{-16}\$](#) and [vs. exact overlap as \$\sigma \rightarrow 0\$](#) ; [full-packing force = \$\nabla\$ energy to \$\sim 10^{-9}\$](#) at $G = 32$; [93 ms/call \(11 \$\times\$ the sharp cost, vs. the Gaussian’s 1040 ms\).](#)

Fully analytic tier (this document; `verify_plummer_analytic.py`, 25-digit working precision, adaptive tanh–sinh references): softened winding (12) vs. 2D convolution, 1×10^{-26} ; primitives (23)–(27), (34), (31) vs. quadrature, $\leq 9 \times 10^{-25}$; pair energy (36) vs. tensor quadrature, 1×10^{-24} ; moments (42), 5×10^{-24} ; total $A_\cap^\sigma$ (40) vs. area quadrature of $\int_A \Psi_B$, 4×10^{-25} ; assembled vertex gradient vs. central FD, 1×10^{-13} (FD-limited); sharp corner formula (17), 9×10^{-26} ; self-repulsion (49), 3×10^{-22} . Sections 7.1–7.3 and 10 are design level; their only numbers are the r_c estimates of (45), arithmetic from the tail formula (44).

References

- [1] H. C. Plummer, “On the problem of distribution in globular star clusters,” *Mon. Not. R. Astron. Soc.* **71**, 460–470 (1911).
- [2] S. J. Aarseth, “Dynamical evolution of clusters of galaxies, I,” *Mon. Not. R. Astron. Soc.* **126**, 223–255 (1963).
- [3] A. J. Chorin, “Numerical study of slightly viscous flow,” *J. Fluid Mech.* **57**, 785–796 (1973).
- [4] R. Krasny, “Desingularization of periodic vortex sheet roll-up,” *J. Comput. Phys.* **65**, 292–313 (1986).
- [5] J. T. Beale and A. Majda, “High order accurate vortex methods with explicit velocity kernels,” *J. Comput. Phys.* **58**, 188–208 (1985).
- [6] K. O. Friedrichs, “The identity of weak and strong extensions of differential operators,” *Trans. Amer. Math. Soc.* **55**, 132–151 (1944).
- [7] L. C. Evans, *Partial Differential Equations*, 2nd ed. (AMS, Providence, 2010), Appendix C (mollifiers).
- [8] J. L. Hess and A. M. O. Smith, “Calculation of potential flow about arbitrary bodies,” *Prog. Aeronaut. Sci.* **8**, 1–138 (1967).
- [9] J. Katz and A. Plotkin, *Low-Speed Aerodynamics*, 2nd ed. (Cambridge Univ. Press, 2001).
- [10] M. Fratanonio and J. J. Rencis, “Exact boundary element integrations for two-dimensional Laplace equation,” *Eng. Anal. Bound. Elem.* **24**, 325–342 (2000).
- [11] A. Carini, M. Diligenti, P. Maranesi, and M. Zanella, “Analytical integrations for two-dimensional elastic analysis by the symmetric Galerkin boundary element method,” *Comput. Mech.* **23**, 308–323 (1999).
- [12] A. Salvadori, “Analytical integrations in 2D BEM elasticity,” *Int. J. Numer. Methods Eng.* **53**, 1695–1719 (2002).
- [13] S. A. Sauter and C. Schwab, *Boundary Element Methods*, Springer Series in Computational Mathematics 39 (Springer, Berlin, 2011).
- [14] L. Lewin, *Polylogarithms and Associated Functions* (North-Holland, New York, 1981).
- [15] NIST Digital Library of Mathematical Functions, §25.12 (dilogarithms), <https://dlmf.nist.gov/25.12>.
- [16] G. H. Golub and J. H. Welsch, “Calculation of Gauss quadrature rules,” *Math. Comp.* **23**, 221–230 (1969).
- [17] M. C. Delfour and J.-P. Zolésio, *Shapes and Geometries: Metrics, Analysis, Differential Calculus, and Optimization*, 2nd ed. (SIAM, Philadelphia, 2011).
- [18] E. Bitzek, P. Koskinen, F. Gähler, M. Moseler, and P. Gumbsch, “Structural relaxation made simple,” *Phys. Rev. Lett.* **97**, 170201 (2006).
- [19] A. Boromand, A. Signoriello, F. Ye, C. S. O’Hern, and M. D. Shattuck, “Jamming of deformable polygons,” *Phys. Rev. Lett.* **121**, 248003 (2018).
- [20] O. Gonzalez and J. H. Maddocks, “Global curvature, thickness, and the ideal shapes of knots,” *Proc. Natl. Acad. Sci. USA* **96**, 4769–4773 (1999).
- [21] C. Yu, H. Schumacher, and K. Crane, “Repulsive curves,” *ACM Trans. Graph.* **40**(2) (2021).

- [22] L. Verlet, “Computer ‘experiments’ on classical fluids. I. Thermodynamical properties of Lennard-Jones molecules,” *Phys. Rev.* **159**, 98–103 (1967).
- [23] M. P. Allen and D. J. Tildesley, *Computer Simulation of Liquids*, 2nd ed. (Oxford Univ. Press, 2017).
- [24] A. W. Lees and S. F. Edwards, “The computer study of transport processes under extreme conditions,” *J. Phys. C* **5**, 1921–1929 (1972).
- [25] H. Wendland, “Piecewise polynomial, positive definite and compactly supported radial functions of minimal degree,” *Adv. Comput. Math.* **4**, 389–396 (1995).